Practical 2 – temperature and pressure gradients

2.1 Calculation of temperature with depth (Data modified after Hirst 2015)

Temperature data have been acquired in 2 wells of the Cheshire Basin. Well locations are in figure 1, and data are in table 1.

1) Calculate a regional geothermal gradient for the Cheshire Basin based on wells A and B, and assuming an average surface temperature of 11.5°C. Exclude data for B1 for now.

2) Later, an extension of well B was carried out by drilling deeper. Temperature data for this well B1 are equally in table 1. What are your observations on the new temperature data, and update your temperature gradient calculations where necessary.

3) How has the geology changed in this well extension?

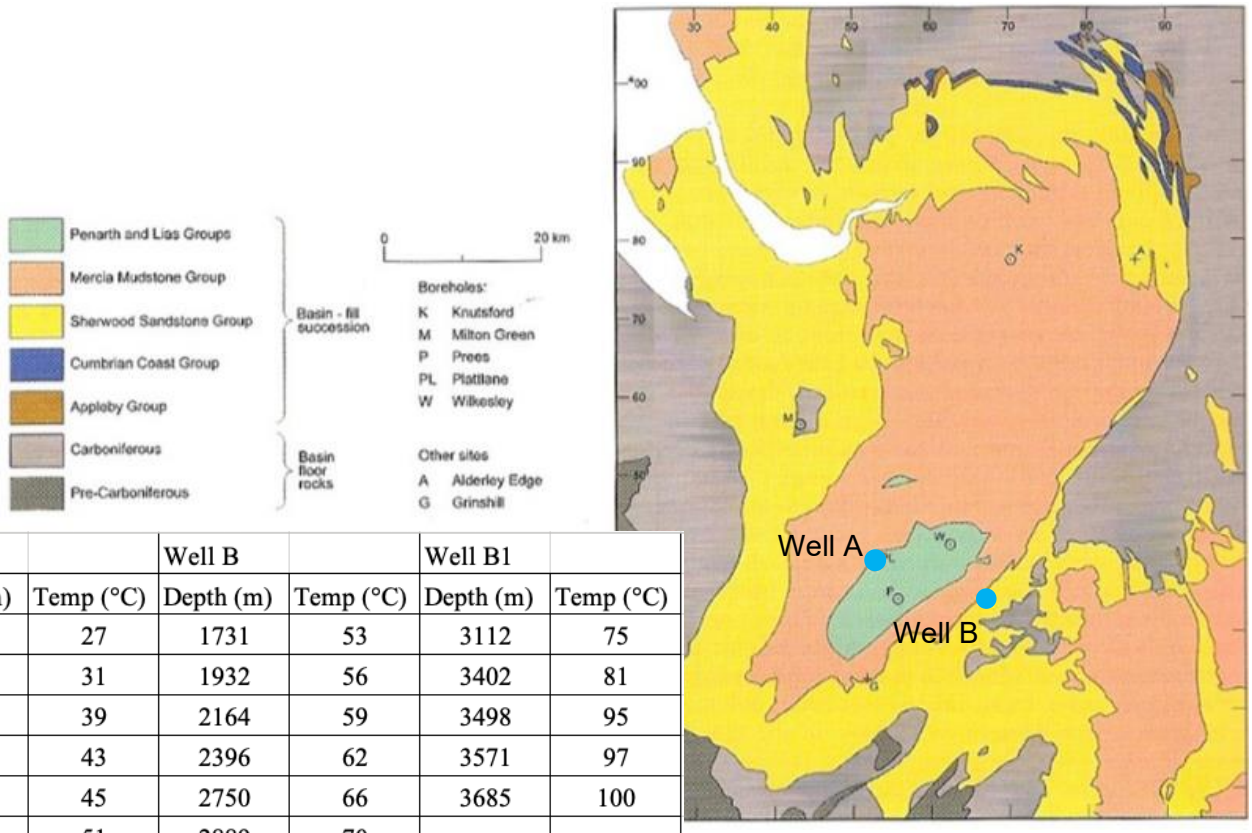


Figure 1: location map

Well A		Well B		Well B1	
Depth (m)	Temp (°C)	Depth (m)	Temp (°C)	Depth (m)	Temp (°C)
715	27	1731	53	3112	75
909	31	1932	56	3402	81
1500	39	2164	59	3498	95
1803	43	2396	62	3571	97
2000	45	2750	66	3685	100
2230	51	2889	70		

Table 1: temperature data

2.2 Calculation of pressure with depth

In the Cheshire Basin a gas well was drilled. The following formation pressures were measured downhole in the well. All depths are in m below ground level.

500m	58 kg/cm^2
1000m	120 kg/cm^2
1500m	175 kg/cm^2
2000m	300 kg/cm^2
2500m	400 kg/cm^2
3000m	450 kg/cm^2
3250m	500 kg/cm^2

Note that values are given in kg/cm^2, which is an old pressure unit, now largely superseded by the SI unit Pa(scal). It is still widely used in drilling though. This unit relates pressure with weight, which is useful in applications where a pressure is applied to generate a load, as in this case.

1 kg/cm^2 = 98,066.5 Pa.

1) Plot formation pressure data against depth.

2) Assuming an average rock density of 2.3g/cm^3, plot the lithostatic (overburden) pressure on the same graph. This is the pressure exerted by the weight of the overlying rock, so is the product of depth and rock density. The value of 2.3g/cm^3 is assumed to be the maximum overpressure that can be encountered before rupture.

Note 1: Mind the conversion!

3) Calculate the hydrostatic pressure and plot it on the same graph.

4) What is the overpressure in the well, i.e. the difference between measured formation pressures and hydrostatic pressures? What does this tell us about conditions in the subsurface?